

ITMS New Server Installation Guide

Repo-native deployment checklist for a fresh Ubuntu host running the full ITMS stack: backend, Postgres, nginx frontend, Salt, Wazuh, ClamAV, and OpenSCAP.

Quick Start

Use this minimal flow when you just want the shortest copy-paste path on a fresh server.

1. Install prerequisites

```
sudo apt-get update
sudo apt-get install -y git curl ca-certificates
curl -fsSL https://deb.nodesource.com/setup_22.x | sudo -E bash -
sudo apt-get install -y nodejs
```

2. Clone the repository

```
git clone https://github.com/Mohdgani-vibe/zerodha-itms.git /home/itms/itms
cd /home/itms/itms
```

3. Run the single installer

```
SERVER_IP=YOUR_SERVER_IP \
+SERVER_NAME=YOUR_DOMAIN_OR_IP \
+DEFAULT_ADMIN_PASSWORD='YOUR_STRONG_ADMIN_PASSWORD' \
+JWT_SECRET='YOUR_LONG_RANDOM_JWT_SECRET' \
+SALT_API_PASSWORD='YOUR_STRONG_SALT_PASSWORD' \
+WAZUH_API_PASSWORD='YOUR_STRONG_WAZUH_PASSWORD' \
+bash docs/install-itms-all-in-one.sh
```

Before You Start

Use a fresh Ubuntu server with outbound internet access and a sudo-capable user account.

- Required before cloning: git, curl, ca-certificates
- Required before frontend deployment: Node.js 22+ and npm
- Installed by the repo scripts: Docker Engine, Docker Compose plugin, nginx, Salt, Wazuh, OpenSCAP packages, and optionally ClamAV
- Open inbound port 80 for the web UI. Use SERVER_NAME when serving by domain instead of raw IP

Seeded Login

For a fresh database, the seeded super admin user is **admin@zerodha.com**.

The password is the value of **DEFAULT_ADMIN_PASSWORD** from **backend/.env.secrets** at first startup.

Any IP Support

The all-in-one installer works on any server IP. It auto-detects the host IP by default and you can override it with SERVER_IP and SERVER_NAME when needed.

```
SERVER_IP=YOUR_SERVER_IP SERVER_NAME=YOUR_DOMAIN_OR_IP bash docs/install-itms-all-in-one.sh
```

Repository

```
sudo apt-get update
sudo apt-get install -y git curl ca-certificates

curl -fsSL https://deb.nodesource.com/setup_22.x | sudo -E bash -
sudo apt-get install -y nodejs

git clone https://github.com/Mohdgani-vibe/zerodha-itms.git /home/itms/itms
```

```
cd /home/itms/itms
```

Single File Install

```
cd /home/itms/itms
SERVER_IP=YOUR_SERVER_IP \
SERVER_NAME=YOUR_DOMAIN_OR_IP \
DEFAULT_ADMIN_PASSWORD='YOUR_STRONG_ADMIN_PASSWORD' \
JWT_SECRET='YOUR_LONG_RANDOM_JWT_SECRET' \
SALT_API_PASSWORD='YOUR_STRONG_SALT_PASSWORD' \
WAZUH_API_PASSWORD='YOUR_STRONG_WAZUH_PASSWORD' \
bash docs/install-itms-all-in-one.sh
```

If SERVER_IP is omitted, the script tries to detect the current host IP automatically.

Config Files To Update

On a new server, only two backend config files must be set before first startup.

- backend/.env: public URL, frontend origin, admin email and name, Salt host, Wazuh host
- backend/.env.secrets: admin password, JWT secret, Salt API password, Wazuh API password

If you use docs/install-itms-all-in-one.sh, the script writes both files for you from the environment variables you pass to it.

Required Files

backend/.env

```
BACKEND_ADDR=:3001
PUBLIC_SERVER_URL=http://YOUR_SERVER_IP
FRONTEND_ORIGIN=http://YOUR_SERVER_IP
DEFAULT_ADMIN_EMAIL=admin@zerodha.com
DEFAULT_ADMIN_NAME=ITMS Admin
SALT_MASTER_HOST=YOUR_SERVER_IP
WAZUH_MANAGER_HOST=YOUR_SERVER_IP
```

backend/.env.secrets

```
DEFAULT_ADMIN_PASSWORD=YOUR_STRONG_ADMIN_PASSWORD
JWT_SECRET=YOUR_LONG_RANDOM_JWT_SECRET
SALT_API_PASSWORD=YOUR_STRONG_SALT_PASSWORD
WAZUH_API_PASSWORD=YOUR_STRONG_WAZUH_PASSWORD
```

Password policy

- At least 12 characters
- At least one uppercase letter
- At least one lowercase letter
- At least one number
- At least one symbol

Installation Commands

1. Start backend and Postgres

```
cd /home/itms/itms
bash scripts/install-docker-and-start-itms.sh --detach
```

2. Verify stack health

```
cd /home/itms/itms
bash scripts/verify-itms-stack.sh --sudo
```

3. Publish frontend behind nginx

```
cd /home/itms/itms
bash scripts/install-itms-nginx.sh YOUR_DOMAIN_OR_IP
bash scripts/smoke-test-itms-nginx.sh --base-url http://YOUR_SERVER_IP
```

4. Install security integrations

```
cd /home/itms/itms
sudo ITMS_ROOT=/home/itms/itms \
  SALT_API_PASSWORD='YOUR_STRONG_SALT_PASSWORD' \
  WAZUH_API_PASSWORD='YOUR_STRONG_WAZUH_PASSWORD' \
  bash scripts/install-itms-server-integrations.sh
```

5. Reload backend configuration

```
cd /home/itms/itms
docker compose -f backend/docker-compose.yml up -d --force-recreate backend
```

6. Install ClamAV for full verification

```
apt-get update
apt-get install -y clamav clamav-daemon || true
```

Validation Commands

```
cd /home/itms/itms
bash scripts/smoke-test-itms-api.sh
bash scripts/verify-itms-security-integrations.sh --wazuh-agent-id 000
```

Expected result: both scripts complete successfully, with SSH checks skipped if the SSH terminal is intentionally not configured.

Admin Password Recovery

If the database already existed and the env password changed later, the admin password must be synced explicitly.

```
cd /home/itms/itms/backend
docker run --rm \
  --network backend_default \
  -v "$PWD":/src \
  -w /src \
  --env-file .env \
  --env-file .env.secrets \
  -e 'DATABASE_URL=postgres://postgres:postgres@postgres:5432/itms?sslmode=disable' \
  golang:1.23 \
  /usr/local/go/bin/go run ./cmd/sync_default_admin_password
```

Expected output: Updated password hash for admin@zerodha.com.

API Login Test

```
curl -sS -X POST http://127.0.0.1:3001/api/auth/login \
  -H 'Content-Type: application/json' \
  --data '{"email":"admin@zerodha.com","password":"YOUR_STRONG_ADMIN_PASSWORD"}'
```

Expected result: a JSON payload with token and a super_admin user object.

Todo Checklist

#	Task	Done when
1	Clone repo into /home/itms/itms	Repo files are present on the server
2	Install git, curl, and Node.js 22+	The host can clone the repo and build the frontend
3	Create backend/.env.secrets	Strong secrets are set before first startup
4	Create backend/.env	Server IP or domain and admin values are set
5	Run Docker installer	Backend and Postgres containers are healthy
6	Deploy nginx frontend	Login page opens on the target IP or domain
7	Install Salt, Wazuh, OpenSCAP	Host services are running and backend env updated
8	Install ClamAV	Security verification can include ClamAV
9	Run smoke test	Core API checks complete successfully
10	Run security verification	Salt, Wazuh, ClamAV, and OpenSCAP verify successfully
11	Log in as admin	UI or API login works for admin@zerodha.com

Generated from the current repo workflow on June 8, 2026. Source scripts: scripts/install-docker-and-start-itms.sh, scripts/install-itms-nginx.sh, scripts/install-itms-server-integrations.sh, scripts/smoke-test-itms-api.sh, scripts/verify-itms-security-integrations.sh.